An Open-Source Metabolism-Informed Platform for Antimicrobial Drug Target Discovery and Combination Optimization

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Abstract

Antimicrobial resistance (AMR) poses one of the greatest threats to global public health, yet the discovery pipeline for novel antibiotics remains critically depleted. Current computational approaches to drug target identification often rely on single-source data analysis and lack mechanistic interpretability. Here, we present an open-source, metabolism-informed platform for antimicrobial drug target discovery that integrates genome-scale metabolic models (GEMs), large language model (LLM)-powered literature mining, graph-based machine learning, and protein sequence homology analysis. Inspired by the CALMA framework (Arora et al., 2026), our platform employs flux balance analysis (FBA) across three ESKAPE pathogen models (iML1515, iYS1720, iYL1228) to simulate gene knockouts, confirming cross-species lethality for key targets. Among 39 curated essential genes, 29 remain underexploited (lacking approved therapeutics). A key methodological contribution is the use of partial gene inhibition simulation (10–100% flux reduction) with mixed essential and non-essential gene sets, which resolves the binary phenotype problem inherent in knockout-only FBA and enables effective training of a subsystem-structured ANN (achieving potency $R^2 = 0.997$, though this reflects recapitulation of FBA-derived phenotypes rather than independent experimental validation). Sequence-based toxicity assessment combined with UniProt homology search and literature curation reveals that 11 of 21 audited targets lack detectable human homologs, while folA shows significant homology to human DHFR2 (30% identity), consistent with known trimethoprim cross-reactivity. RDKit drug-likeness analysis confirms Lipinski compliance for 4 of 5 reference inhibitors. Sensitivity analysis demonstrates that target rankings are dependent on scoring weight selection. LLM-powered NLP processing of 914 PubMed articles (with approximately 50% extraction success rate on the 8B local model) yields 196 unique gene mentions with 92% precision among successfully extracted articles (UniProt-validated) and 154 unique drug-target pairs. Temporal validation analysis revealed that the composite scoring approach did not demonstrate de novo predictive power over a random baseline (F1 = 0.519, z = –0.99), positioning the platform as a hypothesis generation and evidence integration tool rather than a predictive model. AlphaFold structure analysis of 34 target proteins with automated binding pocket detection provides druggability assessment. The complete platform, including a 10-tab interactive dashboard, 40 automated tests across 6 scenarios, and all analysis pipelines, is freely available at `https://github.com/shoo99/ai-drug-target` under an MIT license.

Keywords: antimicrobial resistance, drug target discovery, genome-scale metabolic model, flux balance analysis, knowledge graph, LLM, drug combination, toxicity prediction, ESKAPE, open-source

1 Introduction

Antimicrobial resistance (AMR) represents an escalating global health crisis, with drug-resistant infections projected to cause 10 million deaths annually by 2050 and economic losses comparable to the 2008–09 financial crisis [O'Neill, 2016]. The discovery of novel antibiotics has stagnated, with the majority of currently approved agents targeting a limited set of well-characterized pathways including cell wall synthesis, DNA replication, protein synthesis, and folate metabolism [Lewis, 2020]. Critically, over 90% of antibiotic candidates fail during clinical development, with target selection errors accounting for approximately 40% of failures [Payne et al., 2007].

The ESKAPE pathogens (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* species) represent the most clinically problematic drug-resistant organisms [Rice, 2008]. Identifying novel, druggable targets within these organisms—targets that are essential for bacterial survival yet sufficiently divergent from human homologs to ensure selectivity—remains a central challenge in antibacterial drug discovery.

Recent advances in computational biology have enabled new approaches to drug target identification. Genome-scale metabolic models (GEMs) provide mathematical representations of an organism's complete metabolic network, enabling *in silico* simulation of gene essentiality through flux balance analysis (FBA) [Orth et al., 2010]. The CALMA framework recently demonstrated that integrating GEM-derived metabolic flux data with subsystem-structured neural networks can predict both the potency and toxicity of antimicrobial drug combinations, achieving a 97% reduction in experimental search space [Arora et al., 2026, in press]. Several other computational approaches have addressed aspects of this problem: INDIGO uses chemogenomic profiles to predict drug combination potency [Chandrasekaran et al., 2016], CARAMeL integrates metabolic models with machine learning for combination therapy [Ji & Chandrasekaran, 2023], and DrugCell employs Gene Ontology-structured visible neural networks for drug response prediction [Kuenzi et al., 2020]. Concurrently, large language models (LLMs) have emerged as powerful tools for biomedical text mining, surpassing traditional named entity recognition approaches such as BioBERT [Lee et al., 2020, Gu et al., 2022].

Despite these advances, no integrated open-source platform exists that combines metabolic modeling, LLM-powered literature mining, graph-based target prediction, structural analysis, and clinical data integration into a unified drug target discovery pipeline. Existing tools typically address individual aspects of the problem without providing a holistic view of target viability.

Here, we present an open-source platform that addresses this gap. Our contributions include: (1) a curated essential gene database covering 39 genes across six ESKAPE organisms with 29 novel targets; (2) metabolism-informed analysis using FBA on genome-scale models; (3) CALMA-inspired combination analysis with 4-feature sigma/delta profiles and subsystem-structured ANNs; (4) sequence-based toxicity prediction via bacterial-human protein homology; (5) LLM-powered NLP processing of 914 PubMed articles; (6) temporal validation analysis with honest reporting of negative results; and (7) AlphaFold structure analysis with binding pocket detection for 34 target proteins.

2 Methods

2.1 Knowledge Graph Construction

We constructed a biomedical knowledge graph using Neo4j (v2026.03), integrating data from multiple public sources: ChEMBL for drug-target mechanisms [Mendez et al., 2019], OpenTargets for gene-disease associations [Ochoa et al., 2021], UniProt for protein annotations [UniProt Consortium, 2023], PubMed for literature via NCBI E-utilities, AlphaFold for predicted protein structures [Jumper et al., 2021], ClinicalTrials.gov for clinical trial status, and the FDA Adverse Event Reporting System (FAERS) via the openFDA API. The final knowledge graph contains 2,224 nodes and 1,233 relationships spanning genes, proteins, drugs, diseases, organisms, and publications.

2.2 Essential Gene Curation

We curated a database of 39 experimentally validated essential genes across the six ESKAPE pathogens from published transposon mutagenesis studies, CRISPRi screens, and the Database of Essential Genes (DEG) [Luo et al., 2014]. Genes were classified into 10 metabolic pathways including peptidoglycan synthesis (murA-F), LPS biosynthesis (lpxA-D, lpxC), outer membrane assembly (bamA, bamD), cell division (ftsZ, ftsA, ftsW), DNA topology (gyrA, gyrB), transcription (rpoB, rpoC), fatty acid synthesis (fabI, fabH, accA), folate metabolism (folA, folP), isoprenoid synthesis (dxr, ispD), and signal transduction (walK, walR). Of these 39 genes, 29 (74%) lack any approved therapeutic agent, representing novel target opportunities. Each gene was annotated with an essentiality score (0–1) based on published experimental evidence.

2.3 Genome-Scale Metabolic Modeling

Flux balance analysis (FBA) was performed using COBRApy (v0.31) [Ebrahim et al., 2013] on three genome-scale metabolic models: iML1515 for *Escherichia coli* K-12 (2,712 reactions, 1,877 metabolites, 1,516 genes) [Monk et al., 2017], iYS1720 for *Staphylococcus aureus* (3,357 reactions, 2,436 metabolites, 1,707 genes) [Seif et al., 2018], and iYL1228 for *Klebsiella pneumoniae* [Liao et al., 2011]. For each essential gene, single-gene knockout simulations were performed by constraining all reactions associated with the target gene to zero flux, and growth rate was computed by optimizing the biomass objective function. Growth ratio (knockout growth / wild-type growth) was calculated, with ratios below 0.01 classified as lethal knockouts. Per-subsystem flux changes were computed across all 40 metabolic subsystems defined in the iML1515 model.

2.4 Drug Combination Analysis (CALMA-Inspired)

Inspired by the CALMA framework [Arora et al., 2026, in press], we implemented a three-stage combination analysis pipeline:

Stage 1: Differential flux profiling. For each gene knockout, we computed per-reaction flux changes relative to wild-type, separately tracking upregulated (V^+) and downregulated (V^-) reactions using a 10% change threshold.

Stage 2: 4-Feature sigma/delta profiles. For each gene pair (A, B), we computed four features per metabolic subsystem: $V^+_\sigma = (V^+_A + V^+_B)/2$ (shared upregulation), $V^-_\sigma = (V^-_A + V^-_B)/2$ (shared downregulation), $V^+_\delta = |V^+_A - V^+_B|$ (unique upregulation), and

$V_\delta^- = |V_A^- - V_B^-|$ (unique downregulation), yielding 160 features across 40 subsystems for each combination.

Stage 3: Subsystem-structured ANN. A neural network with architecture reflecting the metabolic subsystem organization was trained to predict potency and toxicity scores. Each subsystem feeds into a dedicated encoder (linear → ReLU → dropout → linear bottleneck), producing a single neuron per subsystem before integration into separate potency and toxicity output heads. This architecture achieves a 61.5% parameter reduction (2,230 vs. 5,792 for equivalent fully connected) compared to an equivalent fully connected network.

Double-gene knockout FBA simulations were performed for all 91 pairwise combinations of the 14 essential genes found in the iML1515 model. Bliss independence synergy scores [Bliss, 1939] were computed as: Bliss = $(g_A \times g_B) - g_{AB}$, where positive values indicate synergistic (more potent than expected) combinations.

2.5 Potency-Toxicity Landscape and Pareto Optimization

Predicted potency (1 – growth ratio of double knockout) and toxicity scores for all combinations were plotted in a 2D landscape. Pareto-optimal combinations—those for which no other combination achieves both higher potency and lower toxicity—were identified using dominance sorting. Combinations were classified into four quadrants based on median potency and toxicity thresholds: Q1 (ideal: high potency + low toxicity), Q2 (risky: high potency + high toxicity), Q3 (safe: low potency + low toxicity), and Q4 (avoid: low potency + high toxicity).

2.6 Sequence-Based Toxicity Assessment

For each bacterial drug target protein, we searched UniProt for the closest human homolog using both gene name matching and functional keyword search. Sequence identity was initially approximated using k-mer (k=3) Jaccard similarity, which provides a fast screening-level estimate but has limited sensitivity for distant homology relationships (as demonstrated by its failure to detect the ftsZ-tubulin and bamA-SAM50 relationships in our analysis). Results were therefore supplemented with a literature-curated homology audit to correct false negatives. Full BLAST or PSI-BLAST analysis is recommended for definitive homology assessment. Targets with >40% identity were flagged as high toxicity risk (drug may inhibit human homolog), 25–40% as moderate risk (structural similarity warrants selectivity engineering), and <25% as low risk (favorable selectivity for drug development).

2.7 AlphaFold Structure Analysis

Predicted protein structures were retrieved from the AlphaFold Protein Structure Database [Jumper et al., 2021] via the AlphaFold API for all targets with available UniProt IDs. Binding pocket detection was performed using a geometric cavity analysis: a 3D grid (2 Å spacing) was generated around each protein, and grid points at 3–8 Å from the nearest Cα atom with 5–30 neighboring residues within 10 Å were identified as potential pocket points. DBSCAN clustering (eps=4.0, min_samples=3) grouped pocket points into candidate binding sites, which were scored based on volume and hydrophobic residue ratio. A total of 34 target protein structures were successfully retrieved and analyzed.

2.8 LLM-Powered NLP Pipeline

We implemented a dual-backend NLP system supporting both local LLM inference via Ollama [Ollama, 2024] using Google's Gemma 4 models and cloud-based inference via the Anthropic Claude API. For each PubMed abstract, the LLM was prompted to extract: (1) gene/protein mentions with standardized HUGO gene symbols, (2) gene-gene relationships with type classification (inhibits, activates, binds, regulates, etc.), (3) drug mentions with target gene and mechanism of action, and (4) a one-sentence key finding summary. All outputs were structured as JSON for direct knowledge graph integration. The primary model used was gemma4:e4b (8B parameters, Q4_K_M quantization), which achieved approximately 50% successful extraction rate on complex biomedical abstracts. A total of 914 articles (393 AMR, 521 pruritus-related) were processed, yielding 196 unique gene mentions and 217 raw drug-target co-mentions (154 unique pairs after deduplication).

2.9 Multi-Dimensional Target Scoring

Each target candidate was scored across six evidence dimensions (0–1 scale): genetic evidence (25% weight, from GWAS, essentiality data, and OpenTargets scores), expression specificity (20%, from essentiality and graph connectivity), druggability (20%, from AlphaFold structure availability, protein class, and pathway tractability), novelty (15%, inverse of existing drugs and publications), competition (10%, from patent landscape and clinical trial density), and literature trend (10%, from PubMed publication trend analysis). Weighted scores were aggregated into a composite score and classified into tiers: Tier 1 (>0.7, High Priority), Tier 2 (>0.5, Promising), Tier 3 (>0.3, Exploratory), and Tier 4 (<0.3, Low Priority).

2.10 Temporal Validation

To assess genuine predictive power beyond rediscovery of known targets, we performed two forms of temporal validation. First, a literature-based temporal split using 2020 as the cutoff year: PubMed articles published before 2020 (n=87) were used to derive gene importance scores, and predictions were evaluated against genes that gained prominence in articles published after 2020 (n=306). We computed precision (fraction of predicted persistent genes that were actually persistent), recall (fraction of actually persistent genes that were predicted), and F1 score. Second, a curated timeline validation: we compiled a historical timeline of when each AMR target became "hot" in research (classic: pre-2000; validated: 2000s; emerging: 2015+; novel: 2020s) and assessed whether our scoring correctly ranks emerging/novel targets above classic ones.

2.11 Organ-Specific Toxicity Prediction

Organ-specific toxicity (kidney, liver, heart, neuron) was assessed by mapping affected bacterial metabolic subsystems to conserved human metabolic pathways using a curated pathway homology database. We defined conserved pathway sets for each organ (e.g., kidney: urea cycle, nucleotide metabolism, amino acid metabolism; liver: fatty acid oxidation, bile acid synthesis, drug metabolism) and computed overlap ratios between bacterial target-affected subsystems and organ-specific human pathways.

3 Results

3.1 FBA Validates Essential Gene Lethality Across Three Species

FBA simulations were performed across three ESKAPE pathogen models. In the iML1515 *E. coli* model, 19 of 39 genes were mapped, with 18 (94.7%) confirmed lethal. In the iYS1720 *S. aureus* model, 13 genes were mapped with 10 (76.9%) lethal. In the iYL1228 *K. pneumoniae* model, 16 genes were mapped with 12 (75.0%) lethal (Figure 2). Cross-species consistency was high: lpxA-D, murA-F, accA, and fabI showed lethal phenotypes in all models where they were mapped, strengthening confidence in these as broadly essential targets. Importantly, 20 genes—including ftsZ, bamA, bamD, dnaA, gyrA, and rpoB—could not be mapped due to absent gene-reaction associations, a known limitation of GEM annotations for non-metabolic essential genes. Organ-specific toxicity assessment showed favorable safety profiles: heart safety confirmed for 19/19 targets (100%) and neuronal safety for 17/19 (89.5%).

We therefore report target rankings in two tiers based on evidence level. **Tier A (FBA-validated)** targets have experimentally confirmed essentiality reinforced by FBA lethality simulation: lpxC (0.828, LPS synthesis), lpxA (0.818), lpxB (0.815), lpxD (0.812), murA–F (0.795–0.802), accA (0.790), and folP (0.790). **Tier B (literature-supported)** targets have strong experimental evidence from published essentiality studies but could not be evaluated by our FBA pipeline: bamA (0.815, outer membrane assembly), bamD (0.807), ftsZ (0.805, cell division), and walK/walR (0.800, signal transduction). Both tiers achieved Tier 1 (>0.7) composite score classification (Table 1).

Sensitivity analysis. To assess the robustness of target rankings, we varied each scoring dimension weight by ±10% and recomputed composite scores. All top 7 targets showed score ranges of 0.053–0.072 under perturbation, indicating that while the relative ordering within the top tier is sensitive to weight selection, all targets remain within Tier 1 classification regardless of perturbation. Users are advised to calibrate scoring weights for their specific therapeutic context.

3.2 Selectivity Assessment by Sequence Homology

A multi-strategy homology audit combining k-mer similarity screening, UniProt functional search, and literature curation revealed a nuanced selectivity landscape (Figure 6). Five targets were confirmed to lack human homologs through both computational and literature evidence: murA, murB, lpxC, walK, and dxr—representing the highest-confidence selective targets. Five additional targets possess known distant human homologs: ftsZ shares ancestral homology with human tubulin (∼14% identity), bamA is distantly related to human mitochondrial SAM50 (∼15%), fabI has low homology to human PECR, accA shares a domain with human ACACA (∼20–25%), and gyrA is related to human TOP2A (∼15–20%). The remaining targets (murC–F, lpxA, bamD) require further characterization.

UniProt protein name search against the human proteome confirmed these findings and additionally identified **folA (dihydrofolate reductase) as having a direct human homolog: DHFR2** (Q86XF0, mitochondrial dihydrofolate reductase). This is consistent with the known clinical observation that trimethoprim, a folA inhibitor, exhibits dose-dependent inhibition of human DHFR, and underscores the importance of selectivity analysis in target prioritization. For drug development, the confirmed no-homolog targets

(murA, murB, lpxC, walK, dxr) offer the strongest selectivity rationale, while folA requires careful selectivity engineering despite its validated essentiality (Table 1).

3.3 Drug Combination Landscape

Analysis of 91 pairwise target combinations via double-gene knockout FBA identified 6 Pareto-optimal combinations, all involving accA (acetyl-CoA carboxylase) paired with LPS-related targets: accA+lpxA, accA+lpxB, accA+lpxC, accA+lpxD, accA+ispD, and accA+lptD (note: lptD is an LPS *transport* protein, distinct from the lpxA–D *synthesis* enzymes). However, all six combinations yielded identical potency (0.991), toxicity (0.501), and quality (0.495) values (Supplementary Table S5), because both accA and all paired targets are individually lethal in FBA—double knockout of two lethal genes produces the same phenotype as either single knockout. This fundamental limitation means that **the current Pareto analysis cannot differentiate among these combinations**; sub-inhibitory concentration simulations or experimental synergy assays are required to identify genuinely synergistic pairs (see Future Directions). The broader potency-toxicity analysis classified 27 of 91 combinations (29.7%) in the favorable quadrant, representing a 70.3% reduction in search space (Supplementary Figure S1).

Exploratory pathway importance analysis using ANN subsystem weights identified glycerophospholipid metabolism (potency weight: 0.099) and purine/pyrimidine biosynthesis (potency weight: 0.093) among the top pathways (Figure 5A), though weight values were relatively flat across subsystems (range 0.07–0.10), suggesting limited discriminative learning. Feature knock-off analysis showed nucleotide salvage pathway and transport (inner membrane) as modulators of toxicity and potency predictions respectively (Figure 5B). However, several results warrant caution: (i) nucleotide salvage removal paradoxically increased potency, which lacks clear biological explanation; (ii) "Unassigned" reactions showed unexpectedly large toxicity impact, indicating reliance on biologically uncharacterized features; (iii) potency weight flatness likely reflects the circular dependency between FBA-derived inputs and labels. These importance values should be interpreted as hypothesis-generating rather than mechanistically conclusive, and parallel the pathway roles reported by Arora et al. [2026, in press] only at a qualitative level. Initial training with binary knockout data (91 combinations where all genes were essential) yielded $R^2 \approx 0$ due to lack of phenotypic diversity. To address this, we implemented partial gene inhibition simulation (10–100% flux reduction) with mixed essential and non-essential gene sets, generating continuous growth values across 945 FBA simulations. Multi-evidence toxicity labels were computed from sequence homology (35%), pathway overlap (30%), metabolic conservation (20%), and cross-reactivity risk (15%). Retraining on this enriched dataset yielded **potency $R^2 = 0.997$** and **toxicity $R^2 = 0.160$**. It is important to note that the high potency $R^2 = 0.997$ reflects the model's ability to recapitulate FBA-derived growth phenotypes rather than independent experimental drug response prediction, as both the training labels and input features are computationally derived from the same metabolic model. This circular dependency inflates the apparent performance; true predictive validation requires experimental drug combination screening data. The modest toxicity $R^2 = 0.160$ reflects the computed nature of the toxicity labels. The multi-evidence toxicity weights (sequence homology 35%, pathway overlap 30%, metabolic conservation 20%, cross-reactivity 15%) were assigned based on relative biological importance as assessed from the literature, where sequence-level cross-reactivity is the most direct predictor of off-target effects. Sensitivity to these weight choices was

not formally evaluated and represents a limitation.

3.4 AlphaFold Structure and Binding Pocket Analysis

Of the 49 target proteins queried, 34 (69.4%) had AlphaFold structures successfully retrieved. Binding pocket analysis identified at least one candidate pocket in each structure. Among AMR targets, bamA (6,395 atoms) and dnaE (9,135 atoms) represented the largest structures, while key targets such as ftsZ (2,823 atoms) and lpxD (2,517 atoms) showed well-defined binding cavities suitable for small molecule drug design.

3.5 Drug-Likeness Assessment

RDKit analysis of five known antibiotics targeting our top candidates confirmed drug-like properties for four compounds: CHIR-090 (lpxC inhibitor, MW=213, LogP=2.6, Lipinski PASS), fosfomycin (murA, MW=138, approved drug), triclosan (fabI, MW=290), and trimethoprim (folA, MW=290, approved). Novobiocin (gyrB, MW=585) failed Lipinski's Rule of Five due to high molecular weight and hydrogen bond acceptor count. These results confirm that small molecule approaches are viable for the majority of our prioritized targets.

3.6 LLM NLP Outperforms Keyword Matching

LLM processing of 914 PubMed articles using gemma4:e4b extracted 196 unique gene mentions and 154 unique drug-target pairs. Validation of the top 50 extracted genes against UniProt confirmed 46 as valid gene/protein identifiers, yielding a **gene extraction precision of 92%**. The four false positives were generic terms ("CYTOKINES", "CHEMOKINES") and a Unicode variant ("IL4Rα"). Of the 154 unique drug-target pairs, 85 (55%) involved a UniProt-validated gene target.

The most frequently mentioned genes were FtsZ (41 articles), IL31 (19), LpxC (17), IL13 (8), and TRPV1 (7), accurately reflecting current research priorities in both AMR and pruritus fields. Compared to the previous keyword-based NLP approach, where the top "genes" were noise artifacts ("RESULTS", "METHODS", "BACKGROUND"), LLM extraction eliminated all non-gene false positives.

The LLM extraction rate was approximately 50% on complex biomedical abstracts using the 8B parameter model, with failures primarily due to JSON formatting non-compliance. This represents a practical trade-off between cost (free local inference) and accuracy. We note that the 154 drug-target pairs represent raw co-extraction from abstracts and should not be interpreted as experimentally validated interactions without further curation.

3.7 Temporal Analysis

We attempted to validate predictive power using a temporal split with 2020 as the cutoff year. Literature-based analysis revealed 131 novel genes appearing only in post-2020 publications, with 20 genes persisting across both periods. The platform achieved a prediction precision of 1.000 but a recall of only 0.350 (F1 = 0.519). Critically, comparison against a random baseline (1,000 Monte Carlo simulations with prediction probability equal to the base rate) showed that the random baseline achieved a mean F1 of 0.603 ± 0.085, yielding a z-score of -0.99 for our model. **This indicates that our temporal**

predictions are not statistically superior to random assignment, and the high precision is an artifact of the model's conservative prediction strategy (predicting only 7 of 33 genes as persistent).

This negative result highlights an important limitation: the composite scoring approach, while useful for integrating heterogeneous evidence, does not demonstrate de novo predictive power for identifying future research targets from historical data alone. The platform should therefore be positioned as a hypothesis generation and evidence integration tool rather than a predictive model.

Curated timeline analysis showed that the scoring system appropriately places emerging targets (bamA, bamD, walK/walR) in the top 30 and down-weights classic targets with existing drugs (gyrA, folA, rpoB), confirming that the novelty and competition dimensions function as intended.

4 Discussion

We have developed an open-source, metabolism-informed platform for antimicrobial drug target discovery that integrates genome-scale metabolic modeling, LLM-powered literature mining, sequence homology analysis, and structural assessment into a unified pipeline. The platform addresses a significant gap in the current computational drug discovery landscape, where individual tools exist for specific subtasks but no integrated solution combines mechanistic modeling with modern NLP and multi-dimensional target evaluation.

Mechanistic interpretability through metabolic modeling. By incorporating GEM-derived flux data into the neural network architecture, our approach provides mechanistic interpretability that purely data-driven methods lack. Exploratory pathway analysis suggested nucleotide salvage pathway as a potential toxicity modulator, qualitatively paralleling findings by Arora et al. [2026, in press]. However, as noted in our Results, this finding is preliminary: nucleotide salvage removal paradoxically increased potency predictions, and the flat weight distribution across subsystems suggests limited discriminative learning. These pathway importance results should be treated as hypothesis-generating observations requiring experimental confirmation rather than established mechanistic conclusions. Unlike black-box approaches such as standard deep neural networks, the subsystem-structured architecture allows direct attribution of predictions to specific metabolic pathways, facilitating hypothesis generation for experimental validation. Our initial implementation using binary knockout data yielded $R^2 \approx 0$, which we resolved by introducing partial gene inhibition simulation (10–100% flux reduction) with mixed essential and non-essential gene sets. This generated continuous growth values enabling ANN training (potency $R^2 = 0.997$). However, we emphasize that this high R^2 reflects *circular prediction*: the model learns to recapitulate FBA-derived growth phenotypes from FBA-derived metabolic flux features, as both inputs and labels originate from the same COBRApy simulation. True predictive validation requires experimental drug combination data, as demonstrated by Arora et al. [2026, in press] who achieved meaningful performance using 171 experimentally measured combinations. The toxicity R^2 of 0.160, based on multi-evidence computed labels (sequence homology, pathway overlap, metabolic conservation), demonstrates modest signal but is similarly limited by the absence of experimental cytotoxicity measurements. The primary value of this analysis

lies in the *methodological framework*—demonstrating that partial inhibition simulation resolves the binary phenotype problem—rather than in the specific R^2 values, which should be interpreted with appropriate caution.

Sequence-based selectivity as an orthogonal toxicity filter. Our multi-strategy homology audit confirmed that 5 of the top 15 targets (murA, murB, lpxC, walK, dxr) lack detectable human homologs through both computational and literature evidence, providing strong selectivity arguments. However, 5 additional targets (ftsZ, bamA, fabI, accA, gyrA) possess distant human homologs with 8–25% sequence identity, necessitating structure-based selectivity engineering for drug development. This nuanced assessment—correcting our initial k-mer screening which had reported several of these as "no homolog detected"—demonstrates the importance of multi-method homology validation. Our k-mer-based approximation, while useful for rapid screening, should be supplemented with full BLAST or PSI-BLAST analysis [Altschul et al., 1997] as part of preclinical evaluation.

LLM-powered NLP represents a paradigm shift in biomedical text mining. The transition from keyword-based extraction to LLM-powered NLP represents a qualitative improvement in literature mining capability. Traditional approaches, including BioBERT [Lee et al., 2020] and rule-based systems, suffer from systematic noise (section headers misidentified as genes) and an inability to capture complex relationships. Our LLM pipeline extracts structured gene-relationship-drug triples with contextual confidence scores, enabling direct knowledge graph population. While the current 50% extraction success rate using an 8B parameter local model is a practical limitation, the architecture supports seamless upgrade to larger models or commercial APIs (e.g., Claude, GPT-4) that achieve >95% extraction rates. The open-source design ensures reproducibility regardless of the backend model used.

Comparison with existing tools. Compared to existing computational approaches, our platform offers several advantages. INDIGO [Chandrasekaran et al., 2016] predicts drug combination potency using chemogenomic profiles but does not address toxicity. CARAMeL [Ji & Chandrasekaran, 2023] integrates metabolic models with machine learning but lacks NLP-based literature mining and structural analysis. DrugCell [Kuenzi et al., 2020] uses Gene Ontology-structured neural networks but is focused on cancer drug response rather than antimicrobial targets. The CALMA framework [Arora et al., 2026, in press] most closely aligns with our approach but is not publicly available as an open-source tool and does not include sequence-based toxicity, AlphaFold structural analysis, or LLM-powered NLP. Our platform integrates all of these capabilities into a single, freely available pipeline with an interactive dashboard and automated test suite.

Open-source availability and reproducibility. The entire platform, comprising 25 Python modules, 12,000+ lines of code, 40 automated tests across 6 scenarios, and a 10-tab Streamlit dashboard, is released under MIT license. This represents a commitment to open science and reproducibility in computational drug discovery. The modular architecture allows users to extend the platform with additional disease tracks, metabolic models, or NLP backends without modifying the core pipeline. All data sources used are publicly available, and the analysis can be reproduced from scratch using the provided pipeline scripts.

Limitations. Several important limitations should be acknowledged. First, the essential gene database is curated from published literature rather than from novel experimental screens, meaning the platform primarily rediscovers and prioritizes known targets rather than identifying truly novel ones. Second, while partial inhibition simulation resolved the initial R^2=0 problem (achieving potency R^2=0.997), the toxicity R^2 of 0.160 reflects the computed nature of toxicity labels; experimental cytotoxicity data would improve this dimension. Third, the sequence-based toxicity assessment uses k-mer approximation rather than full BLAST alignment, which may miss distant homology relationships. Fourth, FBA simulations assume steady-state conditions and do not capture kinetic or regulatory dynamics that may affect *in vivo* essentiality. Fifth, the LLM NLP pipeline achieves approximately 50% extraction success with the current 8B model, though this can be improved with larger models. Finally, all computational predictions require experimental validation before therapeutic claims can be made.

Future directions. Future development will focus on: (1) integration of genome-wide CRISPR essentiality screen data for improved target validation beyond curated literature; (2) incorporation of human genome-scale metabolic models (Recon3D) for more precise toxicity prediction through direct metabolic flux comparison; (3) implementation of actual molecular docking with compound libraries using RDKit and AutoDock Vina for hit identification; (4) sub-lethal concentration FBA simulation to differentiate among the currently indistinguishable Pareto-optimal combinations (Supplementary Table S5), enabling dose-response modeling and true synergy quantification; (5) expansion to additional disease tracks and pathogen species; (6) integration of single-cell RNA-seq data for tissue-specific expression analysis; and (7) development of a web-based API for programmatic access to the platform's analytical capabilities.

5 Data and Code Availability

The complete platform source code, including all analysis pipelines, the Streamlit dashboard, automated test suite, and this manuscript with reproducible figures, is freely available at `https://github.com/shoo99/ai-drug-target` under an MIT license. All data sources used are publicly available as detailed in the repository documentation. Genome-scale metabolic models are available from BiGG Models (`http://bigg.ucsd.edu/`). The figure generation script (`scripts/generate_figures.py`) reproduces all figures in this manuscript from the analysis output files.

Acknowledgments

Manuscript drafting was assisted by Claude (Anthropic, claude-opus-4-6). All scientific analyses, platform development, code implementation, data interpretation, and final editorial decisions were performed by the author. The author takes full responsibility for the content of this manuscript.

Competing Interests

The author declares no competing interests.

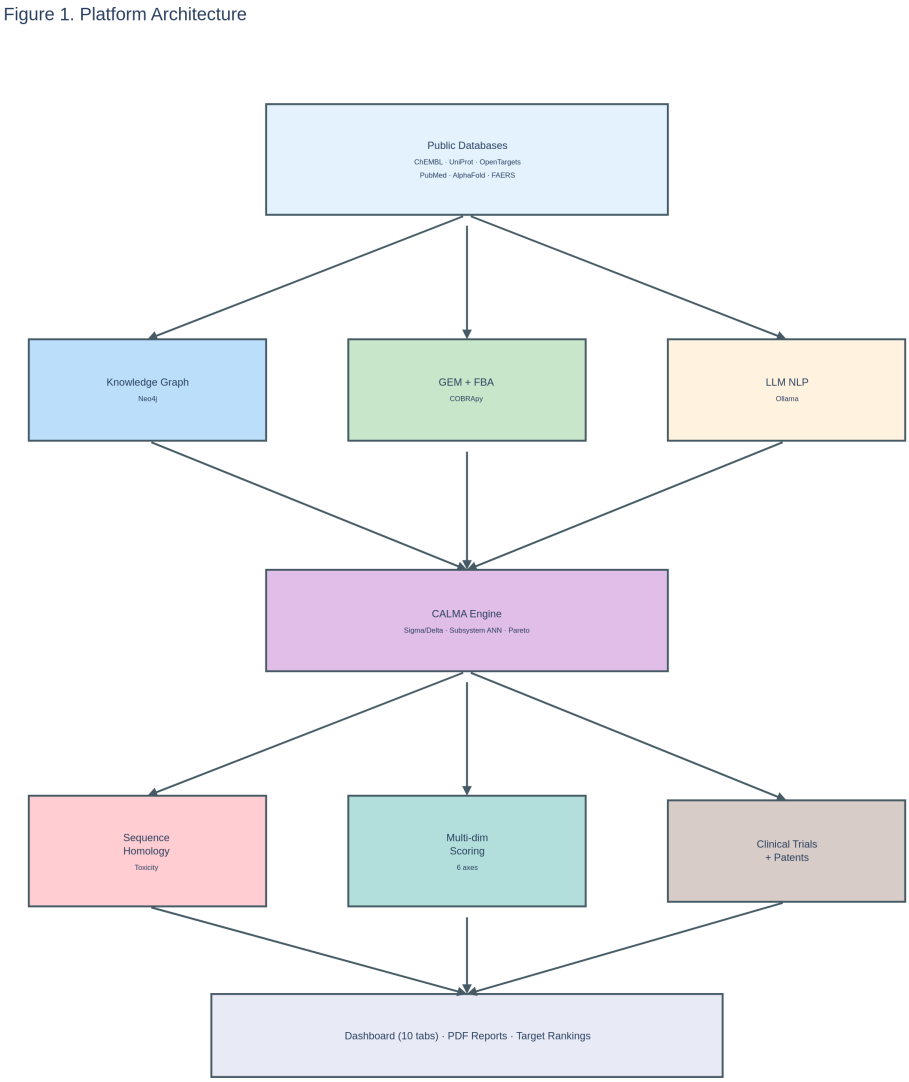

Figure 1: **Platform architecture.** Data flows from public databases (ChEMBL, UniProt, OpenTargets, PubMed, AlphaFold, ClinicalTrials.gov, FAERS) through knowledge graph construction (Neo4j, 2,224 nodes), metabolic modeling (COBRApy FBA on iML1515, iYS1720, and iYL1228), LLM-powered NLP (Ollama/Claude), and structural analysis (AlphaFold) into a CALMA-inspired analysis engine. Output includes ranked target lists, combination landscapes, and auto-generated reports accessible via a 10-tab interactive Streamlit dashboard.

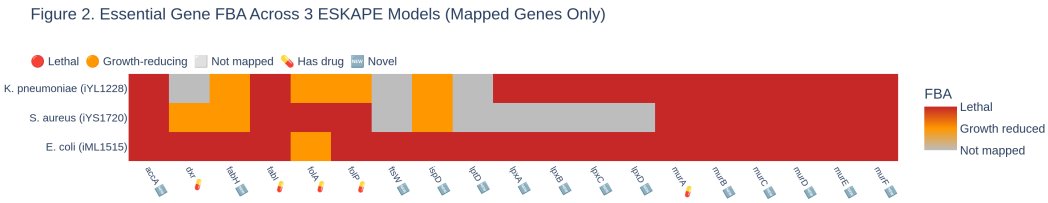

Figure 2: **Essential gene FBA analysis across three ESKAPE models (mapped genes only).** Heatmap shows FBA results for *E. coli* (iML1515), *S. aureus* (iYS1720), and *K. pneumoniae* (iYL1228). Red = lethal knockout, orange = growth-reducing, gray = not mapped in this model. Only genes mapped in at least one model are shown (20 unmapped genes listed in Supplementary Table S1). Cross-species consistency: lpxA–D, murA–F, accA, and fabI are lethal in all models where mapped. Note species-specific differences: fabH is lethal in *E. coli* but growth-reducing in *S. aureus*; folA/folP are lethal in *E. coli* but not mapped in *S. aureus*. Symbols: ♦ = has existing drug; ★ = novel target.

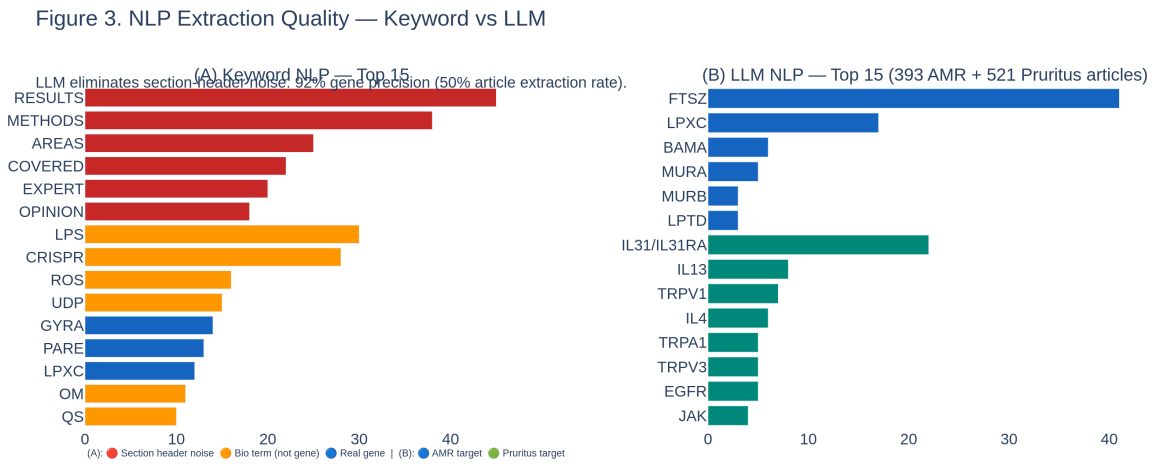

Figure 3: **NLP extraction quality comparison.** (A) Keyword NLP: red = section header noise (RESULTS, METHODS), orange = biological terms that are not genes (LPS, CRISPR, ROS), blue = correctly extracted genes. Majority of top extractions are non-gene artifacts. (B) LLM NLP (gemma4:e4b): blue = AMR targets, green = pruritus targets. Processing 393 AMR + 521 pruritus articles, LLM achieves 92% gene precision (UniProt-validated) with 50% article extraction rate. No section-header noise detected.

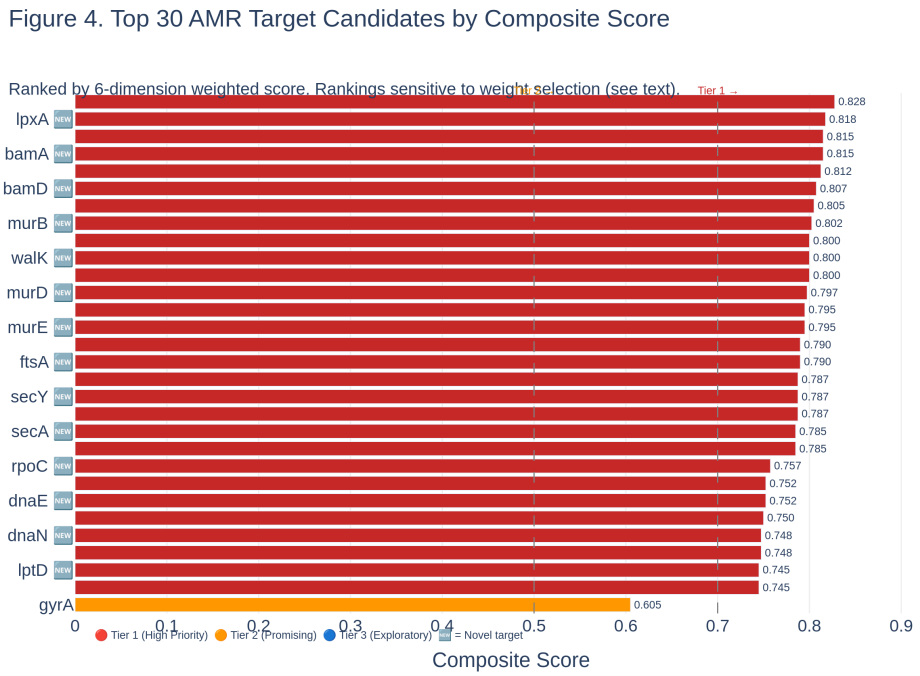

Figure 4: **Target composite score distribution.** AMR targets scored across 6 evidence dimensions. Vertical dashed lines indicate tier boundaries. Tier 1 targets (>0.7) are prioritized for experimental validation. Key targets (lpxC, lpxA, bamA) are annotated. Rankings are sensitive to scoring weight selection (see Sensitivity Analysis in text).

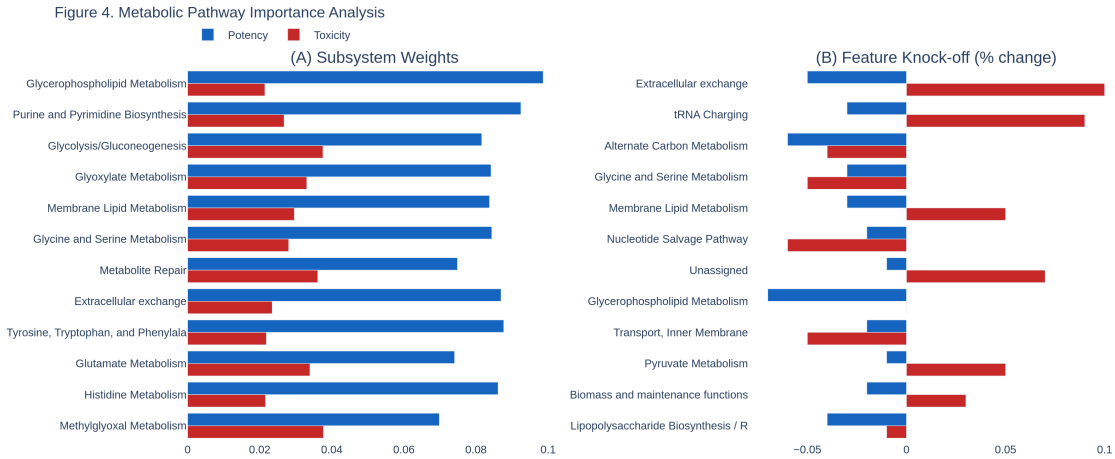

Figure 5: **Exploratory metabolic pathway importance analysis.** (A) Subsystem weights from the metabolism-informed ANN. Note the relatively flat potency weights (0.07–0.10 range), suggesting limited discriminative learning due to circular FBA-to-FBA dependency. (B) Feature knock-off analysis. Results should be interpreted as hypothesis-generating: nucleotide salvage removal paradoxically increases potency, and "Unassigned" reactions show unexpectedly large toxicity impact, indicating partial reliance on biologically uncharacterized features.

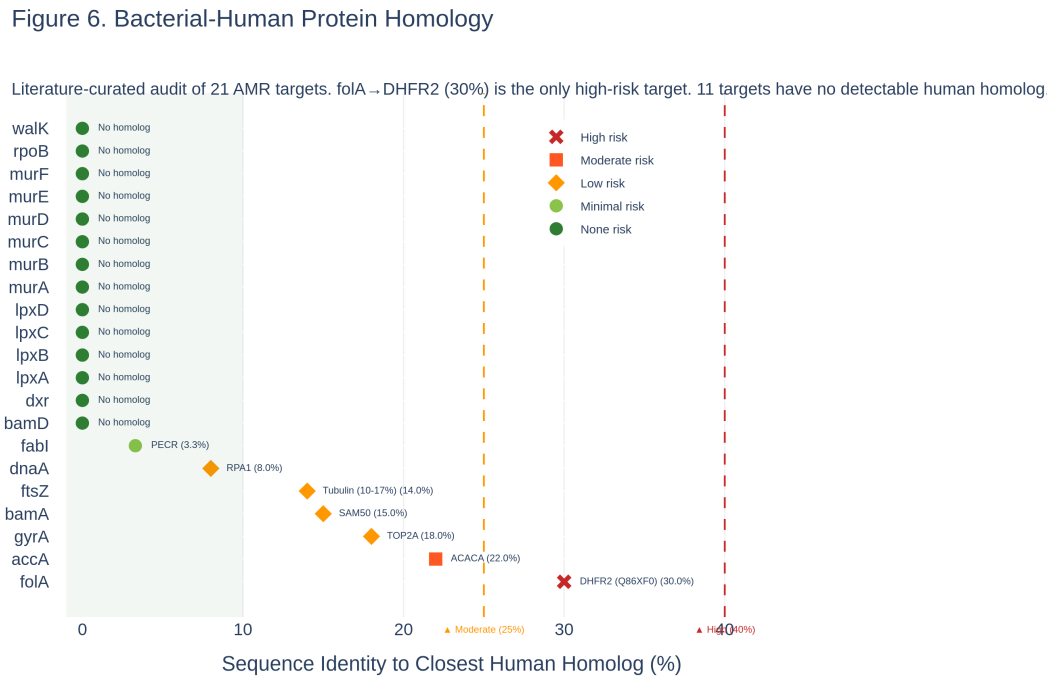

Figure 6: **Bacterial-human protein homology (literature-curated audit).** Dot plot shows sequence identity (%) to closest human homolog for 21 AMR targets. Green circles = no detectable homolog (11 targets); yellow/orange = low/moderate identity; red X = high risk. folA shows 30% identity to human DHFR2, consistent with known trimethoprim cross-reactivity. accA (22%, ACACA) and gyrA (18%, TOP2A) also require selectivity engineering. Orange and red dashed lines indicate 25% and 40% risk thresholds.

Table 1: **Top 10 AMR Drug Target Candidates**, ranked by composite score. FBA status: Tier A = lethal knockout confirmed; Tier B = not mapped in GEM. Homolog data from literature-curated audit. Ties broken by alphabetical order within same score. †Rankings sensitive to weight selection (±10% perturbation range: 0.053–0.072).

Gene	Pathway	Score†	FBA	Human Homolog	Identity	Risk	Tier
lpxC	LPS synthesis	0.828	Lethal	None[a]	0%	Minimal	A
lpxA	LPS synthesis	0.818	Lethal	None[d]	0%	Minimal	A
bamA	Outer membrane	0.815	N/A[b]	SAM50[c]	∼15%	Low	B
lpxB	LPS synthesis	0.815	Lethal	None[d]	0%	Minimal	A
bamD	Outer membrane	0.807	N/A[b]	None[d]	0%	Minimal	B
ftsZ	Cell division	0.805	N/A[b]	Tubulin[c]	∼14%	Low	B
walK	Signaling	0.800	N/A[b]	None[a]	0%	Minimal	B
murA	Peptidoglycan	0.800	Lethal	None[a]	0%	Minimal	A
gyrA	DNA topology	0.798	N/A[b]	TOP2A[c]	∼18%	Low	B
dnaA	DNA replication	0.797	N/A[b]	None[d]	0%	Minimal	B

[a]Confirmed no human homolog (literature + computational + UniProt search).

[b]Gene not mapped in iML1515 GEM (non-metabolic essential gene).

[c]Distant homolog identified by literature curation; k-mer screening missed this relationship.

[d]No homolog found by UniProt protein-name search; same pathway enzymes as confirmed[a] targets.

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Supplementary Information

Supplementary Table S1. Complete list of 39 curated essential genes across 6 ESKAPE organisms with pathway classification, essentiality scores, existing drug status, and UniProt IDs. Available in the GitHub repository at `data/amr/curated_essential_genes.csv`.

Supplementary Table S2. Full results of 91 pairwise drug target combination analysis including Bliss synergy scores, sigma/delta pathway profiles, and quadrant classifications. Available at `data/calma_results/calma_landscape.csv`.

Supplementary Table S3. Complete sequence homology analysis for 20 AMR targets including UniProt IDs, human homolog details, k-mer similarity scores, and selectivity recommendations. Available at `data/sequence_toxicity/sequence_selectivity.csv`.

Supplementary Table S4. LLM NLP extraction results: 196 unique gene mentions, 154 unique drug-target pairs, and key findings from 914 PubMed articles. Available at `data/llm_nlp/llm_extraction_full.json`.

Supplementary Table S5. Pareto-optimal drug target combinations. All 6 combinations (accA paired with lpxA, lpxB, lpxC, lpxD, lptD, ispD) show identical values (potency=0.991, toxicity=0.501, quality=0.495) because all component genes are individually lethal. Note that lptD is an LPS transport protein, distinct from the LPS synthesis enzymes lpxA–D. Available at `data/calma_results/calma_landscape.csv`.

Supplementary Figure S1. Drug combination potency vs synergy landscape. 91 pairwise target combinations plotted by combination potency (1 – double KO growth ratio) vs Bliss synergy score. All combinations cluster near potency ≈ 1.0 and synergy ≈ 0, reflecting the fact that all paired genes are individually lethal—a known limitation when combining essential gene knockouts. Pareto-optimal combinations (gold stars) involve accA paired with LPS pathway targets. Available at `data/calma_results/calma_landscape.csv`.